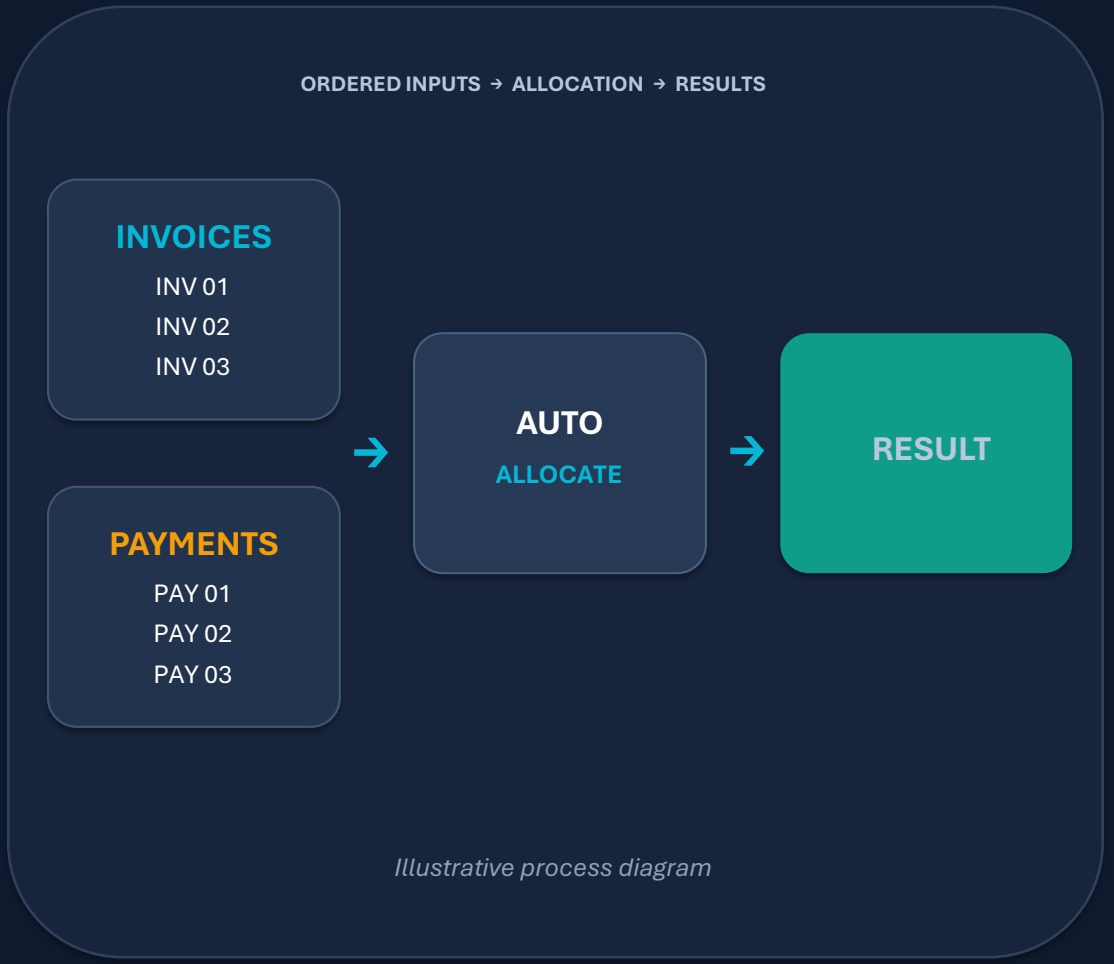
AUTO ALLOCATE

Configurable Payment-to-Invoice Allocation Engine

An Excel-based solution for sequence-driven payment allocation and invoice reconciliation across contractual currencies.

DYNAMIC ARRAY FORMULAS | EXCEL FORMULAS | OPTIONAL POWER QUERY



Developed by Amir Moaied | Senior Project Cost Control Engineer

Payments and invoices rarely settle one-to-one.

Auto Allocate addresses the practical gap between payment amounts and prioritized invoice balances.

Unequal amounts

A payment can be smaller or larger than the remaining balance of the next invoice.

Many-to-many settlement

One payment can fund multiple invoices; one invoice can receive portions from multiple payments.

Priority-driven sequence

Allocation follows the row order supplied for invoices and payments—not a fixed matching rule.

OBJECTIVE

Make sequential allocation repeatable and traceable while allowing business-defined ordering of both input sets.

The engine is designed for allocation—not for choosing the business priority itself.

The engine receives table groups—not one fixed pair.

For every contractual currency in the contract, a corresponding invoice table and payment table are prepared.

EUR **Contractual currency: EUR**

INVOICE TABLE

Rows in selected order

+

PAYMENT TABLE

Rows in selected order

IRR **Contractual currency: IRR**

INVOICE TABLE

Rows in selected order

+

PAYMENT TABLE

Rows in selected order

INPUT PREPARATION

Tables may be arranged manually or prepared with Power Query. Power Query is optional.

KEY PRINCIPLE

The supplied row order defines the allocation sequence. The engine does not impose the sorting policy.

Same sequence-based allocation concept applied within each contractual-currency table group.

Two payments flow across three invoice balances.

Illustrative EUR example: €100,000 and €55,000 are allocated in the pre-set invoice priority order.

01 Read next payment

Take the next payment in the prepared payment sequence.

02 Apply in invoice order

Use the available amount against the next invoice balance.

03 Carry the remainder

Move to the next invoice—or next payment—until funds or invoice balances run out.

PAYMENT QUEUE

Payment 1
€100,000

Payment 2
€55,000

TOTAL: €155,000

INVOICE PRIORITY SEQUENCE

INV-EUR-001	€80,000
INV-EUR-002	€40,000
INV-EUR-003	€40,000

RESULTING ALLOCATION

PAYMENT 1 • €100,000
€80k → INV-001
€20k → INV-002

PAYMENT 2 • €55,000
€20k → INV-002
€35k → INV-003

€155k allocated • €5k open

Payment 1 is split across Invoices 1 and 2. Payment 2 completes Invoice 2 and partially funds Invoice 3, leaving €5,000 outstanding.

Contractual currency sets the allocation basis.

Actual receipt currency and amount remain traceable in the output.

EUR CONTRACT

CONTRACTUAL VALUE

€160,000

ACTUAL RECEIPT

IRR 5.4 billion

Payment ID 2 is included in the EUR payment group.

IRR CONTRACT

CONTRACTUAL VALUE

IRR 2.1 trillion

ACTUAL RECEIPT

EUR 45,000

Payment ID 2 is included in the IRR payment group.

Contractual currency determines the matching table group; actual currency remains a traceable payment attribute.

One chronological result table across contractual currencies.

Selected rows from page 11 of the 11-page Excel output, sorted by invoice dispatch date across EUR and IRR.

Invoice No.	CCY	Dispatch	Line amount	Payment	Received	Actual receipt
INV-IRR-001	IRR	05-Jan-25	1,500,000,000,000	1	05-Mar-25	1,500,000,000,000 IRR
INV-IRR-001	IRR	05-Jan-25	900,000,000,000	2	15-Apr-25	19,286 EUR
INV-EUR-001	EUR	08-Jan-25	100,000	1	05-Mar-25	100,000 EUR
INV-EUR-001	EUR	08-Jan-25	20,000	2	15-Apr-25	675,000,000 IRR
INV-IRR-003	IRR	03-Feb-25	900,000,000,000	—	—	—
INV-IRR-004	IRR	17-Feb-25	1,250,000,000,000	—	—	—
—	EUR	—	15,000	4	18-Jun-25	16,500 USD

Blank payment ID = invoice balance not covered. Blank invoice number = excess payment not allocated.

Line amount represents the allocated portion, remaining invoice balance, or unallocated payment, according to the row references.

The sample tests both sides of the funding balance.

The two contractual-currency groups deliberately use opposite scenarios: excess payment in EUR and insufficient payments in IRR.

EUR CONTRACT

€535,000
Invoice total

€550,000
Contractual payments

100% of invoices allocated

€15,000 contractual payment remains unallocated (actual: $16,500).

IRR CONTRACT

IRR 8.55 tn
Invoice total

IRR 6.40 tn
Contractual payments

74.9% of invoices allocated

IRR 2.15 tn remains: IRR 0.90 tn on INV-IRR-003 and IRR 1.25 tn on INV-IRR-004.

Amounts are shown separately by contractual currency; EUR and IRR totals are not combined.

Excel is the operating environment; allocation logic is the core.

The model separates input preparation from the sequence-based allocation process.

IMPLEMENTATION STACK

CORE LOGIC	**Dynamic Array Formulas** Primary implementation of the sequential allocation.
SUPPORTING LOGIC	**Traditional Excel Formulas** Supporting calculations and checks.
OPTIONAL PREPARATION	**Power Query or manual preparation** Either way, supplied row order is what the engine consumes.

WHAT THE DEMO DEMONSTRATES

- **Sequence-driven**
  Invoice and payment priorities are configurable through input order.
- **Partial and full allocation**
  Amounts flow across invoice balances without exact one-to-one matching.
- **Contractual-currency basis**
  Payment actuals remain traceable while allocation uses contract currency.
- **Unified result**
  Chronological result lines retain invoice balances and excess payment.

Demonstration data is synthetic and was created for portfolio presentation.

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